

ATOMS AND MOLECULES MCQ

1. Matter is defined as any substance that:

- a) only occupies space
- b) has mass only
- c) occupies space and has mass
- d) is visible to naked eye

Correct answer: c

2. The smallest particle of matter visible under standard microscope is:

- a) atom
- b) molecule
- c) electron
- d) collections of atoms

Correct answer: d

3. Ancient atomic theory was proposed by Greek philosophers in:

- a) 5th century AD
- b) 5th century BC
- c) 18th century
- d) 20th century

Correct answer: b

4. Dalton proposed his atomic theory in the year:

- a) 1803
- b) 1897
- c) 1911
- d) 1913

Correct answer: a

5. According to Dalton's theory, atoms are:

- a) divisible
- b) divisible into protons and neutrons
- c) indivisible
- d) created in reactions

Correct answer: c

6. Dalton's postulate that atoms have constant mass was modified due to discovery of:

- a) elements
- b) isotopes
- c) compounds
- d) mixtures

Correct answer: b

7. Modern atomic theory accepts that atoms of same element may have different:

- a) atomic number
- b) atomic mass
- c) charge
- d) size

Correct answer: b

8. Isobars are atoms of different elements having same:

- a) atomic number
- b) mass number
- c) electrons
- d) protons

Correct answer: b

9. Example of isobars given is:

- a) $^{17}\text{Cl}^{35}$ and $^{17}\text{Cl}^{37}$
- b) $^{18}\text{Ar}^{40}$, $^{19}\text{K}^{40}$, $^{20}\text{Ca}^{40}$
- c) H_2 and O_2
- d) Na and Cl

Correct answer: b

10. Atom is defined as the smallest part of an element that:

- a) can be broken chemically
- b) cannot be broken chemically
- c) has no mass
- d) occupies no space

Correct answer: b

11. Atomic structure consists of nucleus containing:

- a) electrons only
- b) protons and electrons
- c) protons and neutrons
- d) neutrons only

Correct answer: c

12. Atomic number Z equals number of:

- a) neutrons
- b) protons or electrons
- c) nucleons
- d) isotopes

Correct answer: b

13. Mass number A is sum of:

- a) protons and electrons

- b) protons and neutrons
- c) electrons and neutrons
- d) protons only

Correct answer: b

14. Number of neutrons $N =$:

- a) $A + Z$
- b) $A - Z$
- c) Z / A
- d) $A \times Z$

Correct answer: b

15. In nuclear notation A_ZX , A is:

- a) subscript
- b) superscript before symbol
- c) atomic number
- d) electron number

Correct answer: b

16. Atomic mass unit (amu) is 1/12th mass of:

- a) hydrogen-1
- b) oxygen-16
- c) carbon-12
- d) helium-4

Correct answer: c

17. Relative atomic mass A_r is ratio to:

- a) hydrogen atom
- b) atomic mass constant
- c) gram atomic mass

d) carbon-12 exactly

Correct answer: b

18. Gram atomic mass of oxygen is:

a) 8 g

b) 16 g

c) 32 g

d) 14 g

Correct answer: b

19. Average atomic mass considers:

a) single isotope

b) isotopes weighted by abundance

c) only common isotope

d) molecular mass

Correct answer: b

20. For carbon, natural abundance C-12 is:

a) 1.10%

b) 98.90%

c) 50%

d) 100%

Correct answer: b

21. Pure substances include:

a) mixtures only

b) elements and compounds

c) only elements

d) only compounds

Correct answer: b

22.Element cannot be broken into simpler substances by:

- a) physical means
- b) chemical means
- c) heating
- d) cooling

Correct answer: b

23.Compound has elements in fixed:

- a) variable ratios
- b) whole number ratios
- c) fractional ratios
- d) no ratios

Correct answer: b

24.Molecule is group of atoms bound by:

- a) physical forces
- b) chemical bonds
- c) magnetic forces
- d) gravitational forces

Correct answer: b

25.Monoatomic molecules are like:

- a) H₂
- b) O₂
- c) He, Ne
- d) CO₂

Correct answer: c

26.Diatomic molecules include:

- a) CH₄

b) H_2 , Cl_2

c) H_2O

d) NH_3

Correct answer: b

27. Triatomic molecules are:

a) H_2

b) CO_2 , H_2O

c) CH_4

d) S_8

Correct answer: b

28. Polyatomic example is:

a) O_2

b) $\text{C}_6\text{H}_{12}\text{O}_6$

c) HCl

d) He

Correct answer: b

29. Homoatomic molecules have:

a) different atoms

b) same type of atoms

c) ions

d) electrons only

Correct answer: b

30. Heteroatomic molecules like:

a) O_3

b) H_2O , CO_2

c) P_4

d) S₈

Correct answer: b

31. Relative molecular mass of H₂SO₄ is:

a) 98

b) 98 u

c) 2+32+64

d) all above

Correct answer: d

32. Mole is amount containing particles equal to atoms in:

a) 12 g C-12

b) 1 g H

c) 16 g O

d) 14 g N

Correct answer: a

33. Avogadro's number is:

a) 6.022×10^{23}

b) 6.023×10^{22}

c) 22.4×10^{23}

d) 1 gram

Correct answer: a

34. One mole of atoms equals:

a) atomic mass in grams

b) molecular mass

c) 22.4 L

d) percentage

Correct answer: a

35. One mole O₂ molecules mass:

- a) 16 g
- b) 32 g
- c) 8 g
- d) 22.4 g

Correct answer: b

36. Molar volume at STP for gas is:

- a) 22.4 L
- b) 11.2 L
- c) 44.8 L
- d) 1 L

Correct answer: a

37. Number of moles = mass / :

- a) atomic mass
- b) molecular mass for molecules
- c) both a and b
- d) Avogadro number

Correct answer: c

38. Percent composition is mass of element / mass of :

- a) atom
- b) compound × 100
- c) mole
- d) isotope

Correct answer: b

39. Avogadro's law: equal volumes of gases at same T,P have equal:

- a) mass

- b) density
- c) molecules
- d) atoms

Correct answer: c

40. In $\text{H}_2 + \text{Cl}_2 \rightarrow 2\text{HCl}$, volumes 1:1:2 implies molecules:

- a) n:n:2n
- b) 1:1:1
- c) 2:2:2
- d) halves combine

Correct answer: a

41. Vapour density is density relative to:

- a) oxygen
- b) air
- c) hydrogen
- d) carbon

Correct answer: c

42. Molecular mass = :

- a) vapour density
- b) $2 \times$ vapour density
- c) half vapour density
- d) atomic mass

Correct answer: b

43. Leucippus and Democritus theory was:

- a) scientific
- b) philosophical
- c) experimental

d) modern

Correct answer: b

44. Dalton's theory based on:

a) philosophy

b) experiments

c) microscope

d) telescope

Correct answer: b

45. Postulate "atoms neither created nor destroyed" from:

a) law of conservation

b) definite proportions

c) multiple proportions

d) all laws

Correct answer: a

46. Modern theory allows mass to energy conversion via:

a) $E=mc^2$

b) fission

c) fusion

d) all

Correct answer: a

47. Isotopes have same Z different:

a) A

b) electrons

c) protons

d) charge

Correct answer: a

48. In lithium nuclear notation ${}^7_3\text{Li}$, neutrons are:

- a) 3
- b) 7
- c) 4
- d) 10

Correct answer: c

49. Electrons mass negligible compared to:

- a) protons
- b) neutrons
- c) protons+neutrons
- d) nucleus

Correct answer: c

50. Gram atomic mass H:

- a) 1 g
- b) 2 g
- c) 1 u
- d) 12 g

Correct answer: a

51. Average atomic mass chlorine considers Cl-35 and:

- a) Cl-36
- b) Cl-37
- c) both
- d) one only

Correct answer: c

52. Mixture differs from compound as elements in:

- a) fixed ratio

- b) variable ratio
- c) chemical bond
- d) same atoms

Correct answer: b

53. All molecules are:

- a) compounds
- b) not necessarily compounds
- c) elements
- d) ions

Correct answer: b

54. Atomicity of P_4 is:

- a) 1
- b) 2
- c) 3
- d) 4

Correct answer: d

55. RMM water H_2O :

- a) 18
- b) 16
- c) 2
- d) 34

Correct answer: a

56. Difference atoms vs molecules: atoms may exist alone, molecules:

- a) always alone
- b) group of atoms
- c) ions

d) electrons

Correct answer: b

57.5 moles O₂ have molecules:

a) 6.022×10^{23}

b) $5 \times 6.022 \times 10^{23}$

c) 10×10^{23}

d) 32×10^{23}

Correct answer: b

58. Moles of NaCl in 29.22 g:

a) 0.5

b) 1

c) 2

d) 58.44

Correct answer: b

59. Gas moles if 44.8 L at STP:

a) 1

b) 2

c) 0.5

d) 22.4

Correct answer: b

60. Atoms in 16 g oxygen atoms:

a) 6.022×10^{23}

b) 3.011×10^{23}

c) 32×10^{23}

d) 1

Correct answer: a

61.% H in H₂O:

- a) 11.11%
- b) 88.89% O
- c) both
- d) 50%

Correct answer: c

62. Gay-Lussac's law explained by:

- a) Dalton
- b) Avogadro
- c) mole
- d) vapour

Correct answer: b

63. Atomicity of gases found using:

- a) Avogadro law
- b) Dalton
- c) mass spec
- d) STP

Correct answer: a

64. Gram molar volume all gases STP:

- a) 22.4 L
- b) from Avogadro
- c) both
- d) 11.2 L

Correct answer: c

65. VD of O₂ is:

- a) 16

b) 32

c) 8

d) 1

Correct answer: b

66. Limitation Dalton: atoms of element have:

a) same mass always

b) different mass isotopes

c) created

d) divisible

Correct answer: b

67. Modern postulate: atoms combine in:

a) any ratio

b) fixed whole number ratios

c) fractions

d) randomly

Correct answer: b

68. Nucleus has protons (positive) and neutrons:

a) negative

b) neutral

c) positive

d) orbiting

Correct answer: b

69. $Z=17$ for Cl, $A=35$ then $N=$

a) 17

b) 52

c) 18

d) $35-17=18$

Correct answer: d

70. Carbon-12 defines amu as exactly:

a) 12 u

b) 1 u for H

c) standard

d) all

Correct answer: a

71. Relative mass no unit as it's:

a) ratio

b) absolute

c) gram

d) kg

Correct answer: a

72. Homoatomic polyatomic:

a) NH_3

b) S_8

c) CO_2

d) H_2O

Correct answer: b

73. One water drop has approx:

a) 10^{21} molecules

b) 10^{10}

c) 1 mole

d) 18 g

Correct answer: a

74. Pen dot has more than:

- a) 2 lakh particles
- b) 100 atoms
- c) 1 mole
- d) visible atoms

Correct answer: a

75. Standard model theory concerns:

- a) planets
- b) subatomic particles interaction
- c) molecules
- d) compounds

Correct answer: b

76. Ion from atom by:

- a) gaining/losing electrons
- b) protons change
- c) neutrons
- d) splitting

Correct answer: a

77. Moles = particles / :

- a) 6.022×10^{23}
- b) mass
- c) volume
- d) density

Correct answer: a

78. For ions, 1 mole NaCl has:

- a) 6.022×10^{23} Na⁺ and Cl⁻

b) molecules

c) atoms only

d) 58.44 ions

Correct answer: a

79. Application Avogadro: molecular formula of:

a) solids

b) gases

c) liquids

d) all

Correct answer: b

80. $E=mc^2$ in modern theory means atom mass can convert to:

a) volume

b) energy

c) particles

d) compounds

Correct answer: b